

Vocal Repertoire of the Squirrel Monkey (*Saimiri sciureus*), its Analysis and Significance

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Summary. Two hundred and fifty vocalizations of the squirrel monkey (*Saimiri sciureus*) were selected for spectrographic analysis from a total of 200 hrs. of tape recordings. The vocalizations were classified into six groups according to their physical characteristics. Both intra and intergroup variability of calls was observed. Calls of similar shape were found to have similar functions. Thus each group of calls could be characterized by a functional designation. The functional significance of calls was determined by qualitative and quantitative observations. Four methods were employed: 1. stereotyped vocalizations were elicited by visual stimuli; 2. motor and vocal reactions were evoked through adequate vocal signals; 3. vocalizations were observed when external conditions were held constant and internal factors were permitted to vary; 4. vocal events were related to the total social situation. By these methods the complexity as well as the specificity of the vocal communication system is demonstrated and its evolutionary significance is discussed.

Key Words: Vocal communication — Primate behavior — *Saimiri sciureus*

Introduction

Our knowledge of sound production in mammals, especially monkeys, is still very scanty. Earlier investigations have led to the assumption that primitive verbal communication in apes appears likely [9, 10, 11, 13, 14, 29]. The findings concerned seem to have been influenced by unwarranted preconceived notions on vocal behavior of such animals. The results of more recent studies no longer support earlier theories [17, 18].

CARPENTER (1934), in the course of studying *Alouatta palliata*, realized the social significance of vocal communication in primates. His statements have been confirmed by several authors [7, 8, 15, 16, 19]. Nevertheless, up to the present time only two species of monkeys have been intensively studied with respect to vocalization, *Macaca mulatta* [23, 24], and the Japanese monkey, *Macaca fuscata* [12].

Although technical difficulties of recording and spectrographic analysis of sounds have been overcome, the problems of the variability of calls and their functional significance still remain. These problems seem to be extremely critical in all high-ranking primates exhibiting an elaborate social group structure. The squirrel monkey is characterized by rich and lively vocalization, even under laboratory conditions [4].

The purpose of this study is to establish a systematically arranged catalogue of the calls of squirrel monkeys based upon their formal characteristics. In addition, an attempt is made to elucidate the function of the calls.

Material and Methods

Maintenance

During the period of investigation — from March to December 1964 — 40 animals of the Brazilian type of *Saimiri sciureus* were available. They were housed in various cages in different rooms:

1. Half of the time (about 100 hours) was spent observing a group of four mature animals, two males and two females. They were kept in an isolation chamber (1.2 m³) adapted for this purpose for over three years, and always seemed healthy.

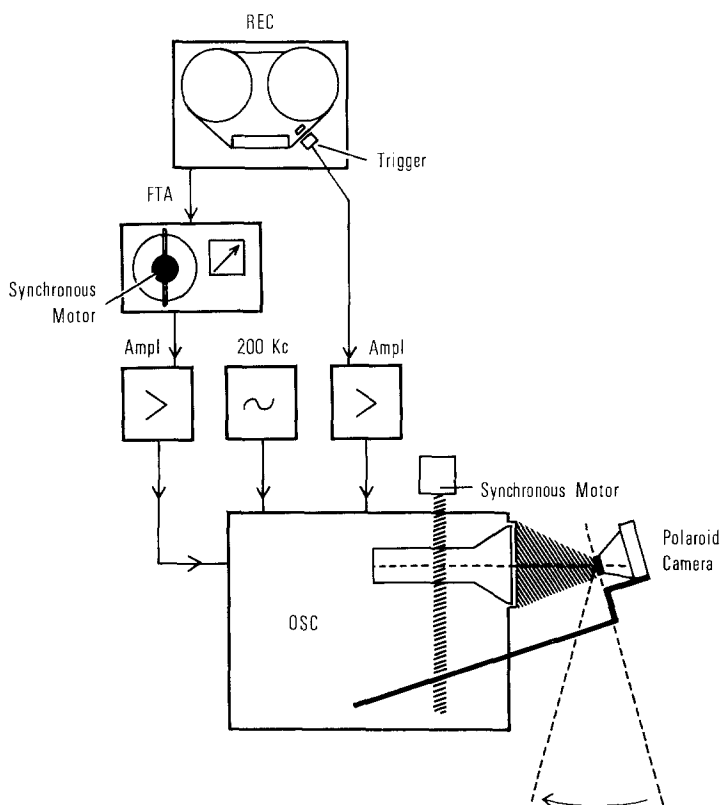


Fig. 1. Block diagram of the apparatus used for sound spectrography according to the method developed by SCHLEIDT (1964)

AMPL = amplifier	OSC = oscilloscope
FTA = Rohde and Schwarz Frequency analyser	REC = Telefunken M 24 tape recorder
	TRIGGER = photodiode

On October 23, 1964 an infant was born, whose growth and development followed a normal course according to our previous experiences [5, 22].

2. A group of six animals, one mature male, two mature females, one immature female, and two young animals were kept in a 20 m³ observation cage. Both groups were used simultaneously for the investigation of social behavior.

3. In order to control specific situations such as feeding and aggression, and in order to obtain high-grade tape recordings, animals were caged individually or in pairs in a sound-proofed room.

4. Additional observations were made in two large collective cages with an outdoor enclosure containing 10—15 animals, which served as a supply for the laboratory.

All cages were provided with a sufficient number of perches. The temperature was kept at 22° C and the day-light period during the winter months was lengthened to 12 hours by additional illumination.

Recording Technique and Sound Spectrography

The following apparatus was used for sound recording: a capacitor microphone (Bruel and Kjaer, 5—20000 cps linear) and Revox-Stereo and Telefunken M 24 tape recorders (17 cps $\pm$ 3 dB, $v = 19.5$ cm/sec). Both were equipped with a device which allowed continuous playing of a call when the tape was formed into a loop. The capacitor microphone, combined with an amplifier, could be used as sound pressure meter. All sound pressures in this paper are expressed in dB re 0.0002 dyne/cm².

Spectrographic analysis was carried out in accordance with the method developed by W.M. SCHLEIDT [26] and modified in this laboratory. The function of the sound spectrograph may be briefly explained as follows (Fig. 1): The loop carries the signal to be analysed. It is read continuously from 0—20 kc by a frequency-analyser (FTA). The amplified (AMPL) output of the analyser controls the intensity modulation of the beam of the oscilloscope. In this way, the brightness of the beam is recorded on the film as differences in gray tones, which represent a dynamic range of 40 dB. Triggering of the time sweep of the oscilloscope with the vocal signal occurs by means of a light barrier, which is activated by the rotating loop. This is accomplished by a photodiode, which is exposed to light when a small, punched hole in the tape runs by immediately before the vocal signal. By continuously tilting a Polaroid Camera so that the successive oscilloscope traces are arranged in vertical sequence, it is possible to record the total frequency spectrum on one picture. When using an analysis range of 200 cps, at least 100 passages of the signal are necessary. The amount of passages was raised to at least twice this quantity in order to improve the quality of the picture.

Characteristic data for each type of call, such as frequency course and duration, were measured, taking into account the number of overtones (Fig. 2). The tape protocol was continuously transferred onto paper in order to facilitate the quantitative analysis of the absolute and relative occurrence of different calls.

The Vocal Repertoire

A. Qualitative Analysis

The recorded sounds were systematically arranged according to their physical characteristics as determined by spectrographic analysis. For the designation of calls, we tried to use expressions that represented characteristic properties of the calls. When we did not find adequate properties, we chose onomatopoeic descriptions. Although we tried to avoid expressions implying functional significance, we used some in relatively clearcut instances.

Terms used by ANDREW [2, 3] and TEMBROCK [25] for distinct types of sounds were referred to wherever possible. Six groups of calls could be distinguished, which are schematically summarized in Fig. 2.

1. Group of peep calls (Fig. 3; Table 1)

These calls are characterized by a uniform frequency pattern, changes in frequency being less than one octave. The fundamental for the majority of these sounds lies above 5 kc. The duration of the calls varies a great deal.

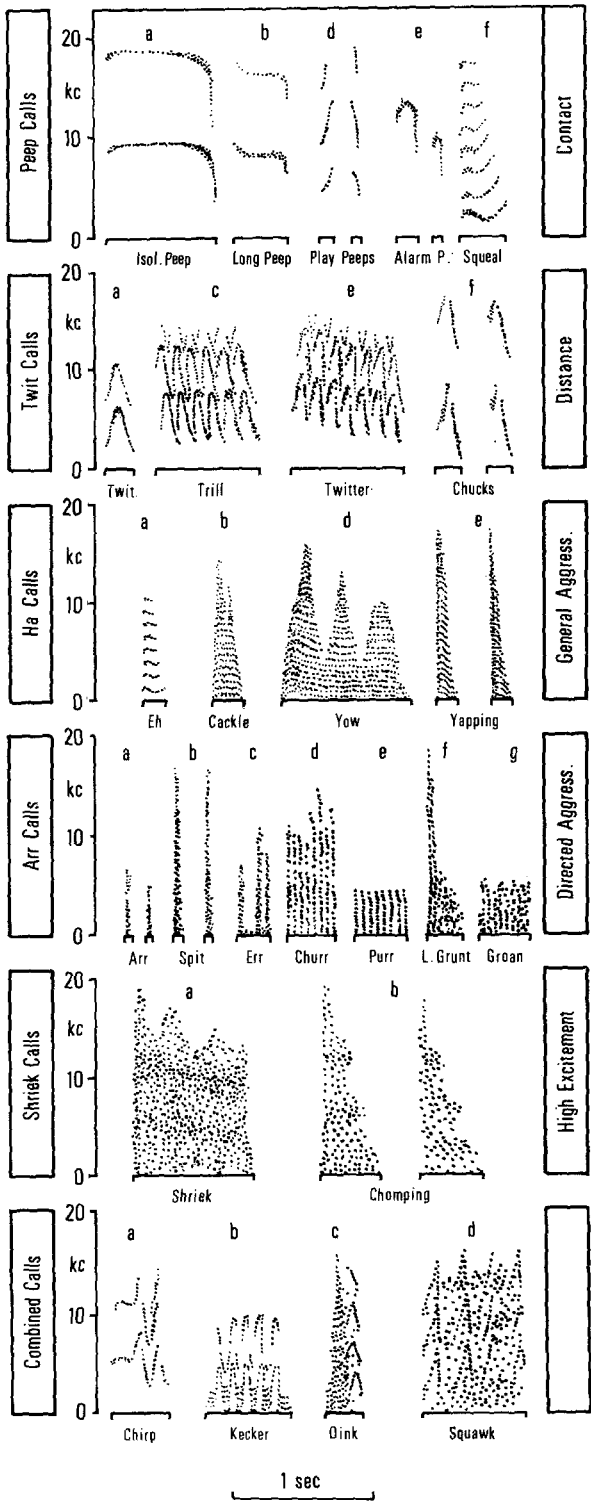


Fig. 2. Schematic presentation of the vocal repertoire of *Saimiri sciureus* according to spectrographic analysis. Letters refer to sound spectrograms of Fig. 3—8

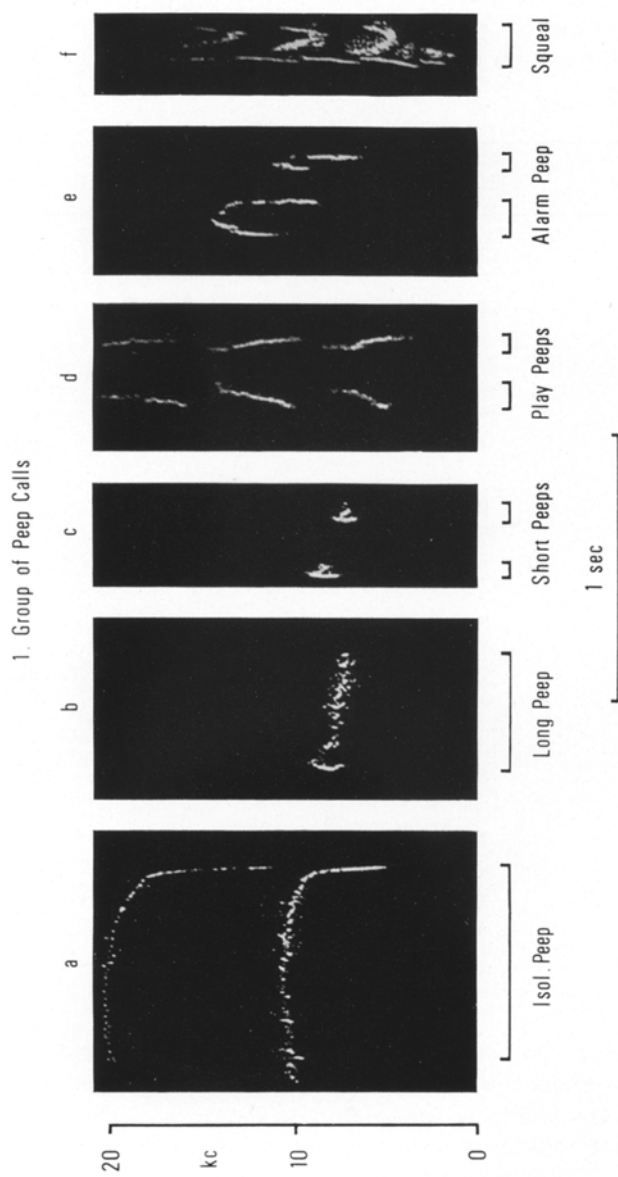


Fig. 3. Sound spectrograms of peep calls

a) Isolation peep (Fig. 3; a)

This stereotyped call is characterized by its high fundamental frequency (9—12 kc), the even course in its central section, and the final steep descent. Compared with the other sounds, its duration is rather long. The sound pressure lies above 120 dB. Isolation peeps are emitted as soon as one or a number of animals loses visual contact with or is separated by a long distance from the group. Hearing this sound, members of the group will answer in the same manner, the

Table 1

call type	n	upper frequency range (kc)	total duration (sec)	duration of single element (sec)
isolation peep	17	9—12	0.7—1.3	
peep	6	7—10.5	0.08—0.69	
play peep	12	5—9	0.03—0.11	
alarm peep	6	11—16	0.05—0.13	
squeal	6			
twit	3	3—7.5	0.1—0.18	
trill	17	6—9		0.09—0.11
twitter	24	8—9.5		0.10—0.16
chuck	16			0.06—0.09
eh	2	0.5—0.8	0.04—0.06	
cackle	46	0.2—0.4	0.06—0.22	
yow	4	0.2—0.4	0.3—0.7	
yap	15			0.1—0.25
arr	3	< 10		0.02
spit	5	> 10	0.02	
err	7	< 15		0.025—0.03
churr	3	> 5		0.03
purr	11	< 5		0.02
labor grunt	10	> 15		0.13—0.2
groan	3	< 5		
shriek	10	> 15	0.3—0.8	
chomp	2	> 5	0.11—0.2	
chirp	5		0.2—0.38	
kecker	7			
oink	3		0.19—0.28	
squawk	7		0.42—0.72	

n = number of calls analysed

result being an exchange of isolation peeps. The following observation may illustrate the degree to which this sound shows almost reflex coupling to separation from the group. When emerging from narcosis, a state is reached in which the animal is able to carry out head movements, whilst the muscles of the trunk and extremities remain uncoordinated. In this state, the stereotyped isolation peep is emitted. It can be elicited easily and continuously merely by repeatedly playing back the sound on a tape recorder.

Evidently, the isolation peep has a seeking function, since it is always emitted as soon as visual contact between one animal and its group is interrupted. The

physical properties, such as long duration, change of frequency, and frequent repetition, are likely to have an important function in acoustical localization. This exchange of isolation peeps may help a separated animal find its way back to the group.

b) Peeps (Fig. 3; b and c)

These sounds differ from the isolation peep by their shorter duration, weaker intensity, and lower fundamental which is on the average about 1—2 kc lower than that of the isolation peep. Furthermore, they exhibit greater variability. Usually there is no pronounced tone plateau. They can take either an upward or downward frequency course. Sometimes, however, peeps can be regarded as short isolation peeps without the drop in frequency at the end.

Peeps are widely used. Peeping has a relatively constant share in the total vocalization scheme in widely varying situations. Exploring animals peep regularly and frequently. They are answered irregularly with the same vocalization by the rest of the group. Persons with neutral or slightly positive valuation receive welcoming peeps; this is also the case when other types of neutral stimulation attract the attention of the animals: for instance when the cage illumination is switched on or off, or as soon as a connecting door is opened. In general, any change of the surrounding stimulus structure, in so far as it does not possess a pronounced positively — or negatively — conditioned feature, elicits peeping.

According to the above indications, peeping is associated with contact and attention. It can be compared with the contact call emitted by some songbirds.

c) Play peep (Fig. 3; d)

Play peeps are characterized by their rapid and sudden increase in frequency. All play scenes seem to be accompanied by play peeping. Since adult animals are seldom seen playing, this call is primarily heard from immature animals. It makes no difference whether playing includes aggressive or sexual components. As soon as grasping, playful nipping, or copulation attempts occur, the frequency of occurrence and the intensity of play peeps are increased. In addition, play peeps may accompany genital display (21), especially in the case of young animals. Play peeps may be elicited by placing a mirror in front of a young animal. The animal will peep while displaying to its own image [20]. Vocalizing in this manner has not yet been observed in fully matured animals.

In spite of some aggressive elements accompanying play scenes, play peeping has never been observed to accompany serious aggression. Genital display combined with play peeps never precedes a fight. These facts seem to indicate, that peeping conveys a non-aggressive intention on part of the emitter.

d) Alarm peep (Fig. 3; e)

The upper frequency range of these very short calls (0.05—0.13 sec) varies between 11—16 kc. Because of the high frequency they may not be heard by an unprepared observer, in spite of the high intensity. The alarm peep was perceived for the first time, when a bird flew across the glass dome of the observation cage. Immediately following the alarm peep, the whole group fled to the highest points of the cage, remaining there for at least 5 sec in a state of motor and vocal inactivity. This reaction has been repeatedly observed since. It has been proved that in addition to birds many other stimuli are capable of eliciting this reaction.

In order to find out if this reaction could be evoked experimentally, a group of three immature animals was caged in a closed room. A disk with a diameter of 20 cm was moved across the surface of the cage at a distance of 1 m and of an angular-velocity of about 5 msec/10°. The complex alarm reaction characterized by alarm peep, flight, and motor and vocal inactivity could be elicited by this stimulus from either side, floor, or ceiling of the cage.

These observations show clearly that this call triggers an alarm reaction, generally elicited when rapid and unexpected stimuli enter into the animals' visual field. The acoustical signal evidently acts as a releaser for the flight reaction, as the latter may be produced by playing back the alarm peeps with a tape recorder.

e) Squeals (Fig. 3; f)

The duration and amplitude of these calls are quite variable. Generally, they are lower in frequency than other peep calls. The course of the fundamental frequency is extremely irregular. This call type is rarely heard; however, it has been recorded in the following situations: by a masturbating mature male, low in rank; by an α -animal (i.e. the dominant animal) separated from but remaining in acoustical contact with the group, in which vigorous sexual intercourse was taking place; by two females during labor [5].

In each of these cases, the other animals in the groups did not seem to react to the vocalization.

ANDREW [3] describes two types of peep calls in squirrel monkeys; one (with average frequency of 8 kc and 0.2—0.3 sec duration) was interpreted as a contact call. This probably corresponds to peeping (s. p. 365). The other type of peep had an initial frequency of 12 kc, which dropped to 8 kc. It is said to be associated with movement and play. Probably it is the same as the play peep (s. p. 365).

On trying to find a common attribute to characterize this group as a whole, taking into account its functions, the most appropriate designation seems to be: group of contact calls.

2. Group of *twit* calls (Fig. 4; Table 1)

The fundamental alternates periodically between 3 and 9.5 kc. The change in frequency is more than one octave.

a) Twit (Fig. 4; a)

This sporadic call with a duration of 0.1—0.18 sec rises and falls between 3000 and 7500 cps. Twits can be considered as being the basic element of the types of sounds referred to below. Due to their rare and irregular occurrence, they could not be associated with specific behavioral situations.

b) Trill (Fig. 4; b, c)

When two or more twit elements are joined to one another with the pitch remaining constant, short (2—3 periods) or long (more than 3 periods) trills result. As in the case of the one-twit element, the duration of one element is about 0.1 sec. The frequency maximum of the fundamental tone lies near the 8000 cps limit. Trilling as well as twittering is a very characteristic call, regularly emitted during feeding. External stimuli, which may be associated with food, also elicit these calls. Another type of behavior is also connected with the trill call: when a

young animal which has temporarily left its mother's back is frightened and flees back, it will utter trilling sounds after having reached the mother's back.

c) Twittering (Fig. 4; d, e)

Twittering is distinguished from trilling by a decrease or sometimes an increase (Fig. 4; e) in pitch. The maximum frequency of the fundamental (9500 cps) is generally higher than that of the average trill. As in the trill, short as well as

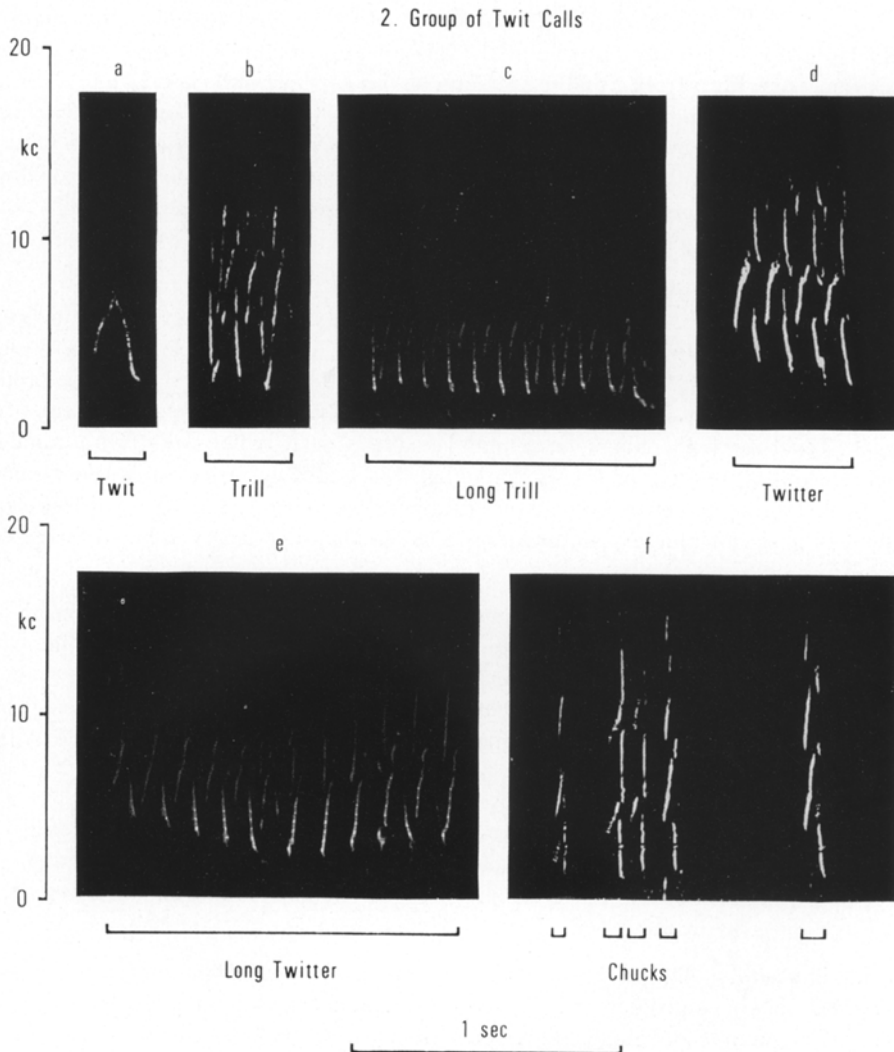


Fig. 4. Sound spectrograms of twit calls

long twitters may be distinguished. Trilling and twittering resemble each other to such an extent that often, in order to distinguish the one form from the other, spectrographic analysis is necessary. A functional relationship probably corre-

sponds to this formal similarity. Both calls are heard during similar behavioral situations, but with different frequencies of occurrence. In addition to the situations in which twittering is heard simultaneously with trilling, the former is also observed in the following situation: an animal which has temporarily been separated from the group will, when put back, be welcomed by twittering calls. The animals react similarly towards familiar persons entering and leaving the room. This call is therefore not restricted to feeding; it has a greeting and announcing function as well.

In an attempt to determine the function of twitters and trills, another aspect may be taken into consideration. Although squirrel monkeys huddle frequently, they usually refrain from huddling as long as one of these calls is emitted. This is especially true in the case of twittering. This behavior is very apparent during feeding, a situation characterized by a high amount of twittering and by a maintenance of minimum individual distances between the animals, varying according to the size of the cage. If these distances are transgressed, aggressive vocalization is elicited.

d) Chuck (Fig. 4; f)

A twit element or a short twitter containing only one part of the up-or downward sweep of the fundamental frequency, but with a steeper descending slope, is designated as a chuck sound. Single and double forms have been registered. Unlike most other calls, a chuck is not likely to be repeated. Instead, it tends to appear together with churrs, arrs, and less often with trills. Huddling animals, when becoming restless or trying to change the huddling order are prone to use this call. It can, however, be registered during other situations. This call presumably has a significance in maintaining a given distance between the animals of a group.

ANDREW (1963) described, without spectrographic illustration, one example of a twittering call with a much lower fundamental frequency (2—5 kc) in which the single elements were of a shorter duration of 0.03 sec. Bearing in mind the spacing-out function in the case of twittering and of place-dominance relationships in the case of chucks, we may designate the calls of this group as distance calls.

3. Group of *ha* calls (Fig. 5; Table 1)

The fundamental frequency is much lower than that of all other calls, and in addition to the fundamental, the sound spectrogram shows a large number of overtones. The frequency course and duration of the individual calls are irregular.

a) Eh (Fig. 5; a)

Eh is a short (0.06 sec), soft call (fundamental about 500—800 cps), which generally occurs singly or rarely in sequences. Sometimes it may be observed in a slightly aggressive context. Its function has not as yet been evaluated.

b) Cackling (Fig. 5; b, c)

Cackling calls belong to the lowest vocalization of squirrel monkeys. The fundamental is below 500 cps. There are a large number of overtones which can be registered up to 16 kc. Duration, as well as intensity, of cackling is variable. Generally, cackling begins with an increase and ends with a decrease in frequency.

The beginning of the cackling call may have a consonant component, too; this means that the acoustical energy is bunched into a narrow band of noise. Due to this fact it is possible to distinguish between two types of cackling: those with and without consonant component. Because of the continuous transitions between these two types, they will be described together. Cackling belongs to those calls which occur in rather varied situations. Cackling regularly occurs in a group of monkeys when a strange animal is brought in. If a new adult male is introduced,

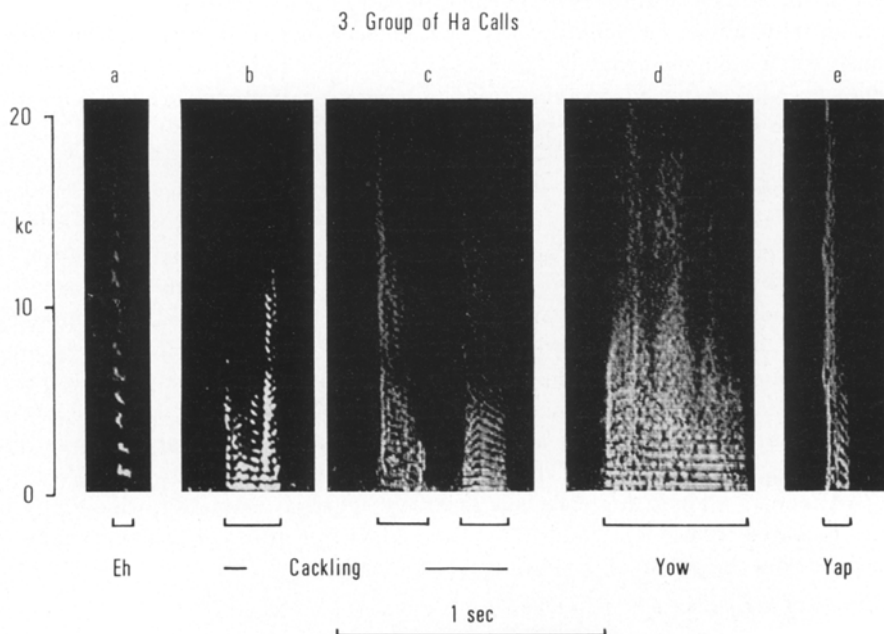


Fig. 5. Sound spectrograms of ha calls

the intensity of cackling will increase until another type of vocalization appears immediately before the outbreak of a fight for rank order. After the fight, when the excitement decreases, cackling appears again. In the same way, strange objects and animals which have no hostile connotation may elicit cackling. Some other situations are: stealing food, occupation and defense of a favorite resting place, unwanted body contact (especially of juveniles with adults), a mother removing her baby from the back.

Cackling is coupled with the initiation of predominantly aggressive behavior and is modulated by the intensity of the stimulus. The range of cackling covers expressions ranging from slight uneasiness to threatening behavior.

e) Yow (Fig. 5; d)

With respect to structure and occurrence, this call is closely related to cackling. It differs from it in its longer duration (0.3—0.7 sec) and a decrease in pitch to the point where it reaches an almost even plateau. In contrast to cackling, yow does not occur in sequences. During violent agitation, for example, before a fight or during the attempt by a mother to push her resisting baby from her back, cackling

may develop into yow. Accordingly, this yow is considered to be the highest intensity step of cackling.

d) Yap (Fig. 5; e)

Yapping may be distinguished by its noise-like initial consonant component. The duration fluctuates between 0.1 and 0.25 sec and there are even more overtones than in cackling. Yapping may be caused simply by presenting a snake or similar object to the group. Animals of other species may also elicit yapping. In general, the animals always react in the same way. As soon as a monkey recognizes a potentially dangerous, moving organism, he flees towards a higher place (if previously sitting on the ground). Simultaneously, by yapping he alarms the group members, who in the same manner seek a higher location and orient themselves towards the suspicious object, yapping.

Intensity and development of this reaction, which is comparable to the mobbing response of songbirds, depend very much on the kind of stimulus. A strong effect can be achieved by furbearing animals. Merely showing a piece of fur to a squirrel monkey may elicit yapping as well as springing to a higher location. It can be distinguished from the alarm peep in two respects: First, the arousing objects do not appear suddenly and unexpectedly in the visual field of the monkey. They have to move rather slowly, whereas objects causing the alarm peep must have a minimum angle velocity. Secondly, a continuous and increasing sequence of vocalizations starts, which may result in aggressive behavior. In consequence, yapping is to be understood as a warning and alarm call which at the same time shows an aggressive capacity. This is not the case with the alarm peep.

Common to all cackling calls is a generally aggressive function which may be communicated from one animal to another. Thus this group of call types may be designated as the group of general aggressive calls.

4. *Group of arr calls (Fig. 6; Table 1)*

Unlike the calls found in groups 1—3, which possess a tonal structure, the sounds of groups 4 and 5 are of noiselike character. The elements consist of single click-like impulses of 0.02 sec duration on the average. One of the characteristics by which the sounds of the group of arr calls may be distinguished from each other is the arrangement of successive signals.

a) Arr (Fig. 6; a)

This short (0.02 sec) pulse-like call has an upper frequency range of approximately 8 kc. It may be detected within aggressive vocalizations such as cackling and spitting. Arr can be regarded as a building stone of all calls included in this group.

b) Spit (Fig. 6; b)

The acoustic energy of this short (0.02 sec) pulse-like noise is distributed almost equally over a frequency range up to 18 kc. Occasionally there follows a second component which is irregularly structured (see labor grunt p. 372). Spitting occurs alone or in connection with head grasping, with the active participant vocalizing. Some other situations are food stealing and genital display. Body contact by means of grasping accompanies the first two of these situations. Obviously we are dealing with an aggressive call directed towards a specific animal.

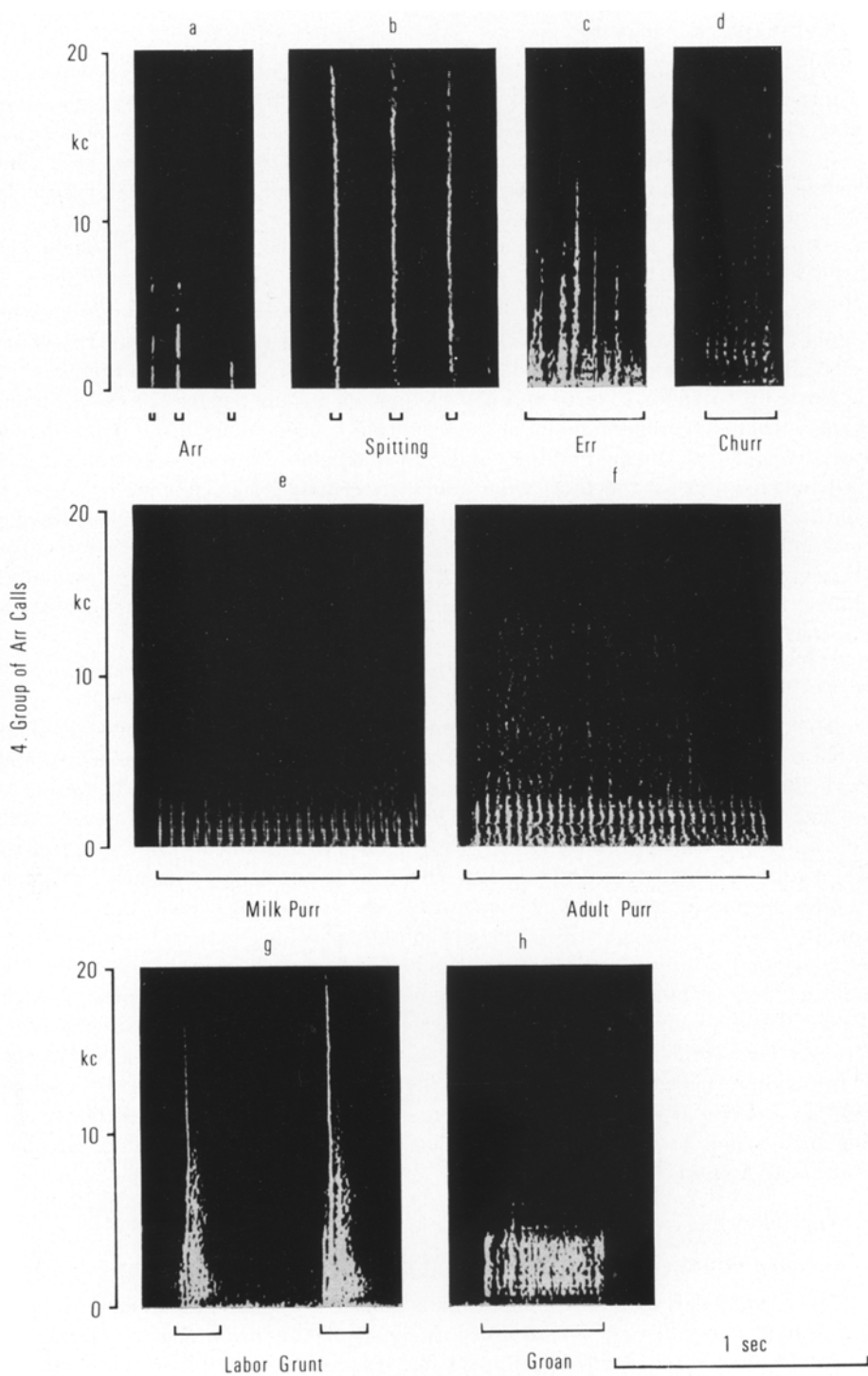


Fig. 6. Sound spectrograms of arr calls

c) Err (Fig. 6; c)

When the arr elements are grouped into short, irregular and blurred sequences of impulses, they are designated as err calls. During an increasingly aggressive motivation, cackling is replaced by err calls. Transitions exist between both types and are made apparent by the consonant component of the cackling call. This change indicates that a general aggressive disposition of an animal will gradually shift into a directed one.

d) Churr (Fig. 6; d)

The essential characteristics distinguishing churr calls from err calls are the regularity and the clear boundaries of single impulses which make up the churr calls (5—9 impulses of about 0.03 sec duration each). Acoustic energy may be detected up to 16 kc. Churrs appear together with other aggressive calls, for instance, when a strange monkey is placed into a group. In addition, it occurs together with chuck during huddling and genital display. Churrs were often uttered by a mother during the first three hours after delivery as, for example, when another monkey tried repeatedly to pull an infant from its mother's back. During these interactions, various aggressive vocalizations were uttered by the mother (Fig. 11; C). In this special case, about 40 % of the total vocalization consisted of churrs. The preliminary conclusion from these observations is that churr has a directed aggressive character.

e) Purr (Fig. 6; e, f)

The number of impulses following in sequence is much greater in purr calls than in churrs. They may exceed 20 in number. The upper limit of frequency of the single impulse is very constant within one call, in contrast to churrs. The impulses are shorter (0.02 sec) and have sharper limitations of frequency than churr calls. The volume is also substantially less. Purring was first noticed in infants prior to the beginning of suckling. Purring and suckling are connected regularly with one another, beginning with the first hours of life, as revealed by observations on four infants [20]. In addition to the purring of infants, which we may call milk-purring, purrs are also uttered by adult animals in connection with genital display, during huddling and within short interruptions of play. Purrs are often associated with churrs. When a female begins smelling or touching an infant on its mother's back, it may utter a series of purrs. Because of the scarcity of the call and its occurrence over a wide range we have only a vague idea concerning the function. In cases of genital display and suckling one can assume that purr indicates a slightly aggressive motivation. The low intensity of this call stresses the fact that it is directed towards an animal close by.

The following calls only appear during the birth process [5]:

f) Labor Grunt (Fig. 6; g)

The formal shape of the labor grunt is similar to spitting, but has a longer duration (0.13—0.2 sec). The frequency range of the beginning impulse may exceed 18 kc. Up till now this call has been heard only during labor pains in the course of birth.

g) Groan (Fig. 6; h)

As in the case of the labor grunt, groans have been registered only during labor pains. In the groan, the intervals of impulses become so short that it is impossible

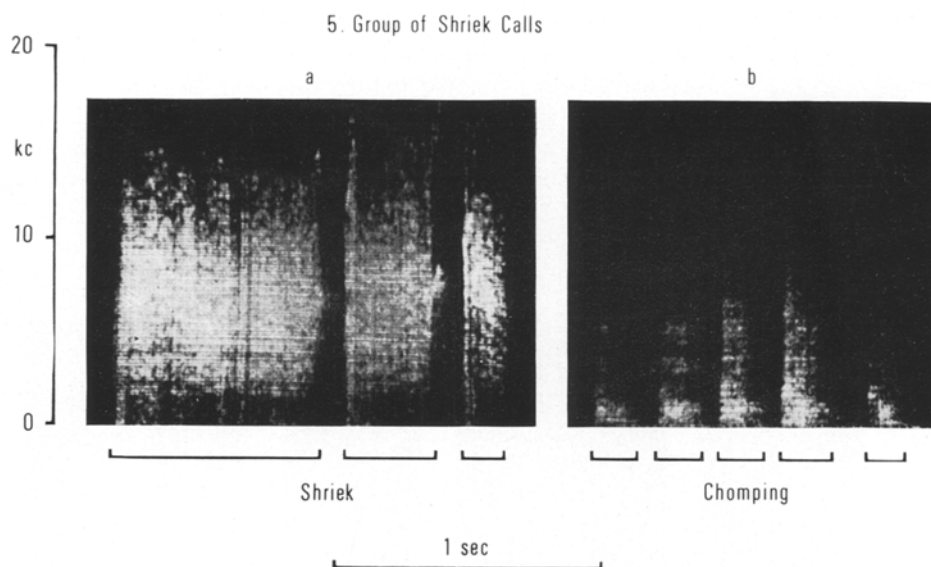


Fig. 7. Sound spectrograms of shriek calls

to differentiate between single impulses. In both cases no reaction from the group could be detected. It is doubtful, therefore, that these calls have any communicative function.

Apart from those calls which have an exclusive connection with labor pains, it is possible to recognize a common function of all arr calls, namely their aggressive and directed nature. Therefore they shall be designated as the group of directed aggressive calls.

5. Group of shriek calls (Fig. 7; Table 1)

These calls are distinguished from the calls of the arr group by a longer duration of at least 0.18 sec to more than one sec. The frequency is distributed over the entire spectrum analysed. In general, a certain frequency is favored and accentuated.

a) Shriek (Fig. 7; a)

The frequency spectrogram covers a range of almost 18 kc. The intensity of the spectrum below 1000 cps is low compared with that between 5 and 7 kc. The range of frequency may vary greatly within one signal. The duration of calls fluctuates between 0.3 and 0.8 sec. Several shrieks usually follow one another, separated by short pauses.

Shrieking is regularly elicited in the following situations: an animal is hurt; a baby is removed from the back of its mother; a juvenile animal is released in a

strange group; an animal is heavily pursued and attacked by another one; two groups of monkeys are put together. In this last case pure "shrieking fights" develop which can lead to fights with severe biting. Shrieking expresses a high degree of excitement, and is almost always transferred to other animals.

b) Chomp (Fig. 7; b)

On the average, the duration of chomping is 0.11—0.15 sec and the call occurs in sequences. The acoustical energy is distributed irregularly up to 12 kc. In contrast to the shriek, the center of intensity is located below 1000 cps. So far, this infrequent vocalization has been heard only from high-ranking males. In several cases it appeared while the group was huddling and the α -male was sitting apart,

6. Group of Combined Calls

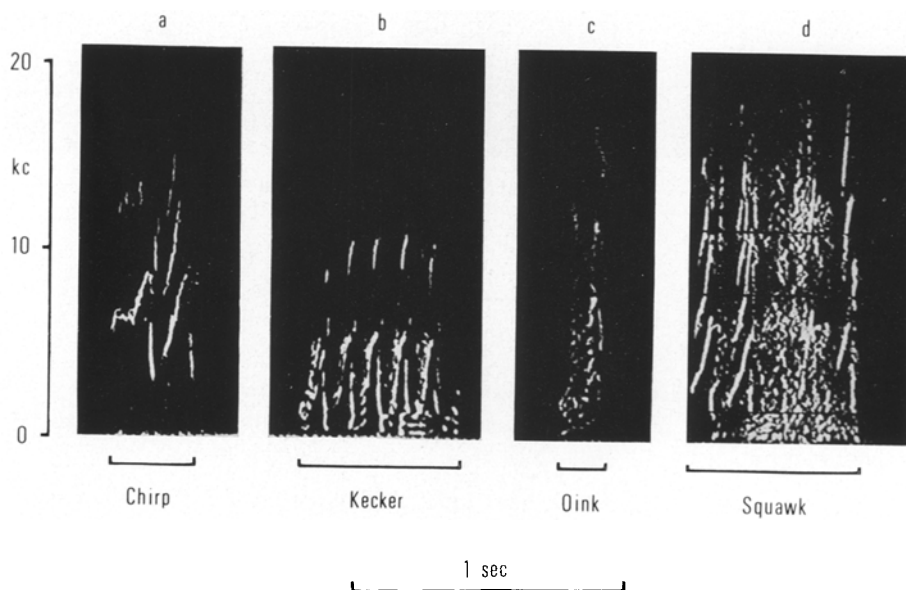


Fig. 8. Sound spectrograms of combined calls

sometimes exhibiting an erection. The spectrogram presented here originates from such a situation. In a further case an α -male began chomping calls when a rival was put into the group. Possibly this call expresses suppressed aggression.

Because of the limited experience with regard to the occurrence of chomping, the chosen group characteristic must principally depend on that of shrieking. The calls of this group will be designated as calls of high excitement.

6. Group of combined calls (Fig. 8; Table 1)

This group is comprised of calls made up of elements of the first to the fifth group and which form a category of transitional calls.

a) Chirp (Fig. 8; a)

This common call consists of a combination of a peep and one or two twits. The duration of the signal is essentially determined by the peep component, the

frequency of which ranges between 7 and 9 kc. This call is usually uttered together with peeps during exploratory behavior or moderate separation of group members. Together with peep calls, chirping belongs to the most common vocalizations of squirrel monkeys, the occurrence of which does not vary so much as other call types. Apparently this call serves the function of a contact call for short to medium distances.

b) Kecker (Fig. 8; b)

Twittering and cackling elements are mixed together in this call. The lower third of the decreasing phase of a twittering period is replaced by a cackling element. Therefore this segment is slightly elongated. If excitation expressed by cackling increases a great deal, the vocalization changes into keckering. In the highest stage of excitation, shrieking occurs. Decrease in excitation is expressed by the inverse order of these vocalizations. Keckers are always released if two groups of monkeys are put together. Like cackling, yapping, and shrieking, keckering has the tendency to transfer a specific mood from one animal to another; thus always more than one animal emits these calls. This causes a high degree of synchronization of behavior patterns, which would be advantageous during aggressive interactions of groups.

c) Oink (Fig. 8; c)

A cackling call leading into a twit or a peep of increasing frequency is defined as an oink. In our experience, this call is limited to animals under one year of age. It seems to replace the milk-purr in babies older than three months. It is also uttered by babies if they wish to regain their mother's back. Oinks may also occur if the baby displays general excitement. Clearcut oinks were never heard from adults.

d) Squawk (Fig. 8; d)

The squawk is a noise-like call of 0.57 sec mean duration which is interwoven with twits. If an animal tries to steal food from another, a squawk may be elicited from one of the animals involved. Similar reactions may result when two or more animals quarrel for a preferred sitting place. It is often preceded by cackling, indicating an increase in aggression.

7. Supplementary list of non-vocal sounds

a) Sneezing

Sneezing is a relatively frequent behavior pattern. The frequency of its occurrence varies with different situations. It is rather common that when one animal starts sneezing others will follow, so that one may observe bursts of sneezing.

b) Gnashing of teeth

Several times in the course of minor surgical operations under local anesthesia we observed gnashing of teeth. So far we have never heard this sound from animals within the communal situation.

c) Shaking

In the state of high excitement adult and particularly young males like to jump on objects and shake them intensely thus producing a loud noise. This is a behavior pattern which has already been described for other species of primates, such as the shaking of trees and drumming on hollow trunks of trees [1, 12].

As was shown above, it was possible in only a few cases to determine convincingly the function of calls on the basis of mere qualitative observations. In many cases we cannot see any simple and immediate reactions following an acoustic signal. Therefore, we have prepared a synopsis of situations in which different calls

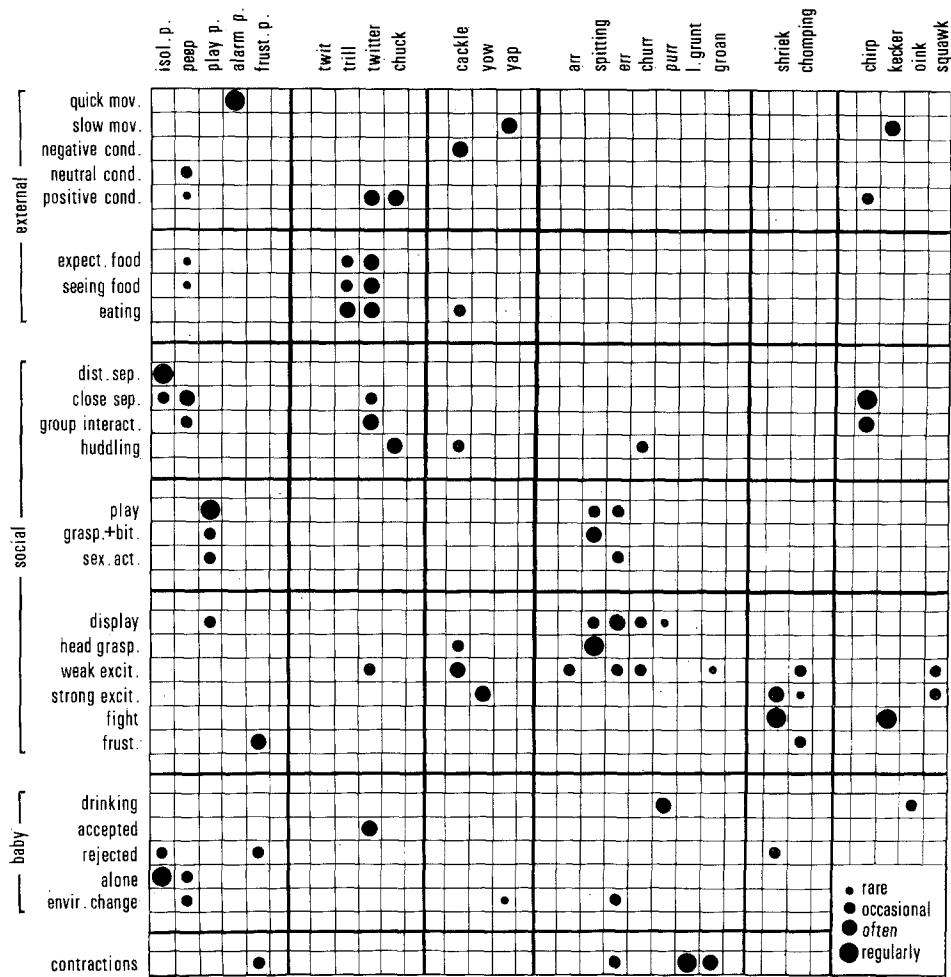


Fig. 9. Characteristic distribution of call types in different situations. Horizontal axis: call types arranged in groups. Vertical axis: situations in which vocalizations occur. Diameter of circles indicate the frequency of occurrence of calls in a certain situation as estimated by qualitative observation

- Abbreviations:
- act. = activity
bit. = biting
cond. = conditioned
dist. = distant
frust. = frustration/squeal

envir. = environmental
excit. = excitation
expect. = expecting
grasp. = grasping
iso. = isolation

l. = labor
mov. = movement
p. = peep
sep. = separation
sex. = sexual

are likely to occur (Fig. 9), in which the six groups of calls are plotted on the horizontal axis and the different situations during which they may occur on the vertical axis. The amount of interaction is indicated by circles of different diameters: in the case of some call types, there are clearcut relationships between specific

situations and specific calls, while in others we find a broad dispersion. However, there exists a definite accumulation of points representing call types of one group in related behavioral situations. For instance, contact with calls of the group of peeps, feeding with trills and twitters, aggressive behavior with the group of arr calls.

B. Quantitative Findings

Qualitative observations on the functional significance of calls are summed up in Fig. 9. It shows that in some cases one call type can be found in different situations involving different stimuli. The animal's reaction to a particular stimulus often consists of a complex sequence of vocalizations.

In Fig. 10; A, B for example the degree of motivation was changed by food deprivation. The behavioral protocol reports that in B there is a significant increase of aggressive behavior such as stealing food, pushing apart and persecuting. The highest level of aggressive motivation can be observed in C, after a strange male has been put into the cage. In each of the three situations a certain amount of contact calls persists with peeping and chirping, occupying between 20 % and 40 % of the total vocalization. Fig. 10; A represents a distribution of frequencies typical for the feeding situation, characterized by a predominance of twittering and trilling calls. Only about 10 % of the total vocalization consists of cackling calls. After 30 hours of food deprivation (Fig. 10; B), the distribution is altered a great deal. The amount of twittering and cackling has changed as compared with A. In addition, keckering and shrieking are recorded. When a strange male animal is introduced (Fig. 10; C), there are no longer any twittering or trilling calls.

The percentage of cackles has decreased in favor of churrs and purrs. This happens in connection with an increase of directed aggressive actions such as genital display

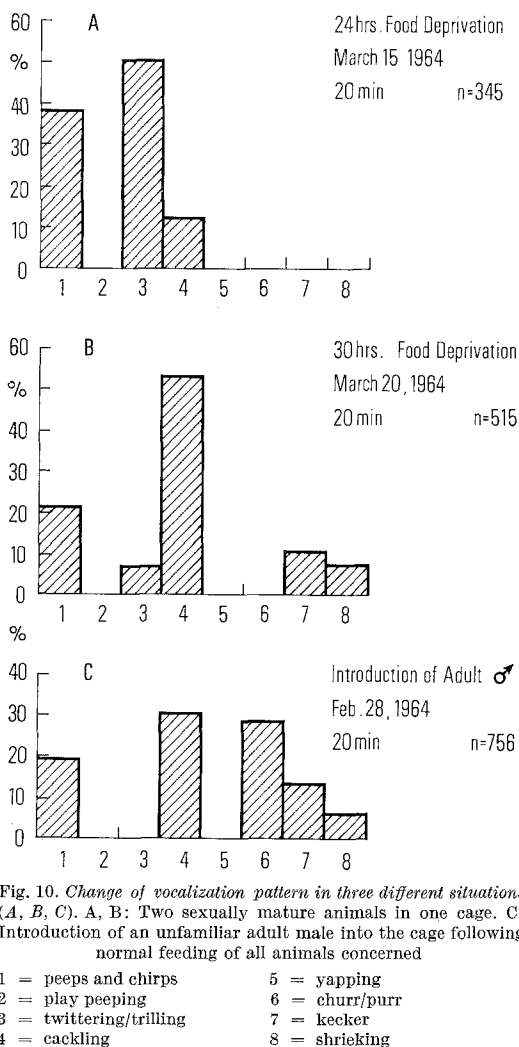
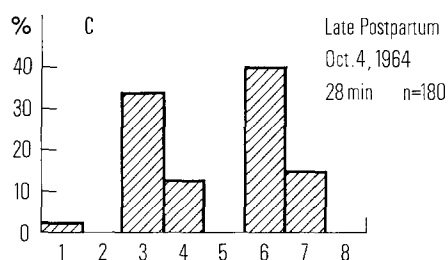
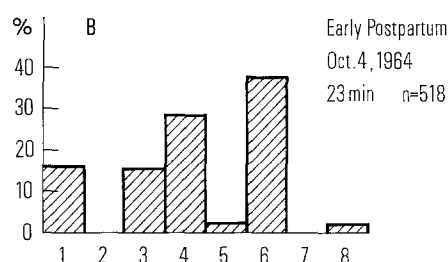
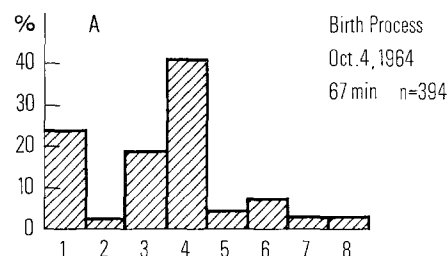


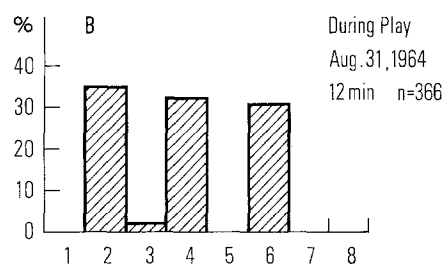
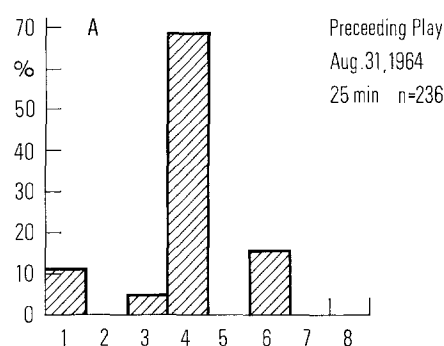
Fig. 10. Change of vocalization pattern in three different situations (A, B, C). A, B: Two sexually mature animals in one cage. C: Introduction of an unfamiliar adult male into the cage following normal feeding of all animals concerned

[20], persecuting, biting and fighting. Another example of increasing aggressive motivation as revealed by behavioral observation is demonstrated in Fig. 11; A, B, C. It shows the vocalization within three stages during the course of a birth [5]. The increase in directed aggression is paralleled by a relative decrease in



←
Fig. 11. Change of vocalization in a group of four animals during delivery. Most vocal exchanges between mother and premature female. Call types according to Fig. 10

Fig. 12. Change of vocalization pattern in a group of four monkeys before and during play. Call types 1—8 according to Fig. 10
↓



cackling, peeping and chirping and an increase in churrs and purrs during the late postpartum period. The relatively high amount of twittering in the postpartum periods is due to feeding.

These data confirm the concept of cackling calls as being an expression of a general aggressive tendency, whereas churr and purr calls are connected with directed aggressive actions.

Another example from a different motivational system is added in Fig. 12; A, B. In this case, a spontaneously occurring change in motivation was observed and analysed. Before the onset of play a relatively high amount of cackling was recorded. This general aggressive tendency (A) suddenly changed (B) and play of all four animals in this group developed. As directed aggressive elements of behavior, such as grasping and sham biting, increased, so did the intensity and the amount of churrs and purrs. Additionally, a new call type characteristic for this situation appeared: play peeping, which is supposed to signalize the play character

of these aggressive actions. For further elucidation, these data have to be investigated according to sequential relationships.

Discussion

Twenty-six different call types could be distinguished by means of spectrographic analysis. These are divided into five groups. The sixth group is comprised of calls formed by elements of one or more of the five groups. It is not claimed that this catalogue covers the complete vocal repertoire present in the natural habitat. Nevertheless, we have good reason to believe that the essential part can be observed under optimal laboratory conditions. Groups varying with respect to size, sex ratio, age, past history and housing conditions show almost the same vocal repertoire. This was confirmed by observations in other laboratories [21].

For the systematic arrangement of calls in a catalogue we used criteria derived from spectrograms: time and frequency course of the fundamental, number of harmonics, intensity and patterning of the signal. When compared to a functional arrangement of vocalization, this typological one has some important advantages [17, 18], viz.

1. The criteria defining the call types can be formulated objectively.
2. The function of a call does not have to be taken into consideration at the beginning of the study.

In order to establish a catalogue of vocalizations according to physical characteristics by comparing spectrograms, a great number of spectrographic analyses must be performed. However, this method fails when the differences between physical characteristics are too minute to be picked up by the spectrogram. In the course of this study, 250 spectrographic analyses have been performed. They were selected from about 200 hrs of tape recordings: later, they were repeatedly compared with other tape recordings. Samples showed that this selection, used as a basis for comparison, proved to be representative. Nevertheless, subjective estimations could not be entirely eliminated. This is due to the variability of call types which is a difficult problem in all studies of primate vocalization [24, 27]. No two spectrograms are ever exactly identical. In general, two types of variability must be taken into account. One is concerned with variations in duration, volume and pitch of calls within one group of calls. It may be called intragroup variability. The second is concerned with transitions between calls of different groups. It may be called intergroup variability. There are no difficulties in comparing spectrograms of calls exhibiting intragroup variability. Calls characterized by intergroup variability, however, cannot be classified by spectrographic analysis alone. An example may illustrate this problem (Fig. 13).

Calls of the third and fourth group are characterized by a high degree of both types of variation. By decreasing the pitch of the eh call, cackling may develop. On the other hand cackling may be transformed into yow by elongation and into yapping by shortening and increase in intensity. By the insertion of consonant elements transitional calls are formed leading to the group of arr calls (Fig. 13). For understanding this type of call, functional analysis must also be used, in addition to spectrographic analysis. Thus, before progressing to discuss details of vocal variabilities, the function of a call has to be defined: a specific call or a sequence of calls transmits information from one animal to another, in this way

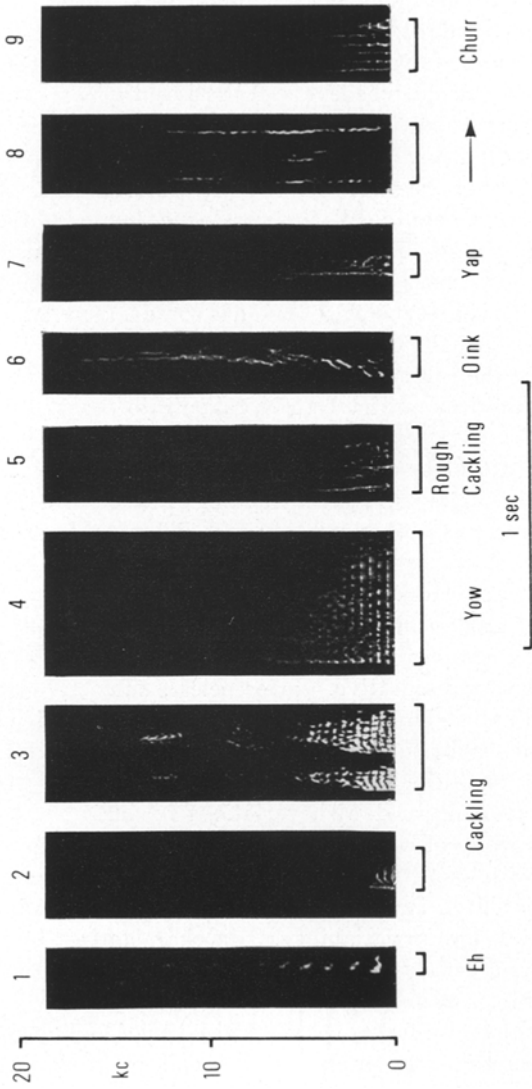


Fig. 13. *Variability of call types.* 1—4 intragroup variation in pitch and duration; 5—7 intergroup variation, appearance of arr elements in ha calls; 8 transitional call type

influencing their behavior. The change in behavior evoked by an acoustic stimulus reveals its function. The calls described serve a social function, aside from a few exceptions which are mentioned in the text. A squirrel monkey isolated from members of its own species will cease vocalizing within a few days.

Four methods can be employed to delineate the information content of these calls:

1. Repeated elicitation by well defined visual stimuli;
2. Elicitation of specific behavior patterns (vocal or motor) by playing back tape recordings of the vocal patterns recorded from the animals;
3. Observation of vocal behavior while varying the animals' motive state;
4. Correlating types of calls with the social role and status of the animal (emitters and receivers) within the group.

1. In order to get a preliminary idea of the information content of a vocal event it is sufficient to present a defined visual stimulus (situation) and to register the vocalization following the stimulus. In an experiment of this type, the visually perceived situation evokes a clearcut stereotyped reaction which can be simply vocal, or vocal and motor.

Isolation peeping (s. p. 364), alarm peeping (p. 365), twittering (p. 366) or yapping (p. 370) are examples of vocalizations recorded in this type of experiment. In these cases, the visual stimulus is adequate for the release of a specific vocal response. But the possibilities of this method of functional analysis are limited. First of all only a small part of the vocal repertoire can be elicited in this way. Further, all calls listed above will cease if the stimulus is presented too often in succession. Under certain circumstances, some calls, yapping for instance, can never be elicited if the same stimulus is used repeatedly (p. 370). The call will appear again if another strange one is presented.

2. The second method uses the vocal response itself as an adequate acoustic stimulus. It is replayed by a tape recorder and projected into the cage by a loud-speaker. The reaction of the animals will be either vocal (such as the isolation peep) or motor (for example flight to the highest point of the cage after presentation of an alarm peep) or a combination of vocal and motor (for instance, after presentation of yapping, flight from the ground to some elevated point, aggressive attitude and yapping). After repeated use, these acoustic stimuli remain effective for longer periods than visual stimuli and the imperative character of the acoustic signal seems to be stronger than the visual one. Only a small number of calls and sequences can be elicited in this simple way, however. A record of shrieking recorded during a fight will never trigger a real fight. The animals will become excited, but will not begin fighting.

3. By means of these two methods of functional analysis, different external situations are used as a stimulus for checking whether a particular type of vocal behavior is specific or not. A third method can be used, however, in which the external factors are kept constant and the internal ones are varied. The technique employed was food deprivation for different time intervals. Qualitative differences could easily be detected. But only a quantitative analysis including all calls gives a convincing result. In such an experiment, we no longer deal with specific calls correlated to specific stimuli, but rather, the entire vocalization pattern changes in a characteristic way. In one example (Fig. 10) the pattern of the vocal behavior

changes in relation to the time spent since the last feeding; the amount of aggressive vocalization increases with increasing hunger. This result seems to be self-evident. Nevertheless, it is important for functional analysis because it shows that apparent coincidence of functions within groups of calls as revealed by the different methods we employed is not the result of an arbitrary arrangement but an expression of specific motive states.

Aggression and playing are other topics for motivational analysis (Fig. 11; 12) indicating that internal factors have to be considered. The effect of these factors can be assessed in terms of corresponding changes in frequencies of occurrence of different call types. When considering play- and aggressive behavior, any simple distinction of external stimulus and internal motivation is impossible, because the characteristic pattern of vocalization will only appear in social situations: an appropriate partner must be available. The complex interaction of vocal, tactile, visual, and possibly olfactory stimuli is a prerequisite for the development of the total repertoire of calls. This is the reason why the playback of calls is not sufficient for functional analysis. In many cases, it is necessary to present a visual stimulus simultaneously in order to get a vocal response.

As vocal partnerships can be manipulated only within narrow limits, behavioral observation of groups of monkeys is still the most important method for functional analysis. Upon reduction of the number of animals in a group, not only motor, but also vocal behavior is substantially reduced. Some calls can be heard only if at least four animals of different ages, including both sexes, form the group. Other calls are heard only in the presence of babies and young animals, as was shown in the example of the oink call (s. p. 375).

4. The fourth and final method for functional analysis of calls consists in analyzing the social behavior of groups containing animals of different age and sex. This is an issue which has concerned this laboratory for several years [20]. Experience gained during this time can be used for analyzing vocal behavior. The results show that the functional significance of a call can sometimes be understood only in relation to its social context. The particular reaction evoked by vocalization is dependent not only on the specific information content of the vocalization but also on the social rank of the sender and the emitter.

The reaction of a mother to an infant's purring varies according to the age of the infant. During play, peeping is uttered only by the partners engaged in playing, not by other members of the group. Thus, in many cases, vocal responses are specific to the current social status of an animal. This fact has to be taken into account in any functional analysis.

The provisional results of our studies of the function of calls according to the four methods mentioned permit us to conclude that physically similar calls do show some similarity in communicative functions. It is for that reason that we have chosen for each group of calls not only a descriptive term but also an additional designation which is intended to characterize the function of this group of calls. Whether all call types within one group do fit into this scheme cannot be decided at this time. Thus far, this functional grouping appears to work. In further refinements of this functional analysis, some modifications of the scheme may become necessary. Another more sophisticated method will be the study of transient probabilities of vocalization periods [5].

The call types of the first two groups are relatively invariant and stereotyped, whereas types of the third and fourth group show a rather high degree of variability. Calls of the first two groups are chiefly concerned with the spatial distribution of the members of the group. Additional visual stimuli are of subordinate value. The calls of the third and fourth group appear with close social contact, together with visual stimuli. The reason for this difference in variability may be that calls which are supposed to function without visual stimuli have to be stereotyped, whereas calls used in close social interactions of partners may require a higher degree of variability. "Misunderstandings" are excluded by the visually guided social context in which a call occurs.

This hypothesis can be supported by comparing diurnal with nocturnal animals, as for instance *Saimiri sciureus* and the systematically related *Aotus trivirgatus*. From studies of MOYNIHAN [19], one gets the impression that the vocal repertoire of *Aotus* is much less variable than that of the squirrel monkey.

The few call types of the fifth group are relatively uniform. They are of little value in the study of the vocal communication system. Loud shrieking results in transmitting excitement from one animal to another.

The sixth group is a problematical one for functional analysis, because the calls composing this group appear to be combinations of calls from two or more of the other groups. The main question is, whether these calls resemble a summation of single elements from other groups of calls, or whether they are signals in themselves, transferring specific information.

The extensive repertoire of squirrel monkeys includes the whole scale of vocal elements so far known in primates. In this highly developed communication system, which is paralleled by the social behavior of these animals, the question of the evolution of such a system arises.

First of all, the adaption of squirrel monkeys to an arboreal way of life has to be taken into account. It is important for social animals that the coherence of the group be guaranteed, even within a biotope difficult to survey. Peep and twitter calls seem to be suitable for this purpose because of their physical properties and little variability. Ha and arr calls contribute considerably to the formation of group structure. Their greater variability corresponds to the more complicated social interactions in connection with which they occur. From an evolutionary standpoint, it seems to be appropriate to limit the range of variability of calls in order to eliminate possible vocal interference of different sympatric species. Which of these factors are of determinative significance must be learned from comparative studies.

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